

ALAN WRYA

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Science (BSc), Financial Mathematics Co-op | Computer Science Minor

Sept 2022 – June 2026

- **CGPA: 3.95** | Dean's List Scholar (all eligible terms)
- *Relevant Coursework:* Stochastic Calculus, Partial Differential Equations, Investment Analysis, Managerial Finance, Probability & Statistics, Linear Algebra, Differential Equations, Data Structures, Computer Science

EXPERIENCE

TD Bank — Direct Investing

Jan 2025 – Aug 2025

Co-op, Direct Investing Journey Analyst

Toronto, ON

- Rebuilt 12+ monthly KR/OKR dashboards in Excel using **VBA**, LAMBDA functions, dynamic named ranges, and reusable chart templates — cutting manual refresh time from 2–5 hours per week to under 5 minutes per tab.
- Engineered automated 24-month rolling performance charts with dynamic labels, $\pm 3\sigma$ control limits, and auto-scaling axes, enabling leadership to refresh reports on demand and standardize reporting across the team.
- Built client segmentation frameworks (Core Funded, Newly Funded, Reactivated, Attrited) and conducted correlation/regression analysis on 3 years of monthly client data to identify drivers of retention, funding, and engagement.
- Led analytics tagging plans for new Direct Investing features — reviewed wireframes, scoped high-value events, presented to tagging/analytics teams, and coordinated risk sign-off across product, analytics, and 1B controls.
- Conducted feature-launch lookback analyses (e.g. enhanced search adoption, mobile vs. desktop trading behaviour) in Adobe Analytics and Excel to inform roadmap prioritization and executive-facing recommendations.
- Took on expanded analytics, stakeholder, and reporting responsibilities during the manager's parental leave.

Ted Rogers Sales & Trading (TRST)

Sept 2023 – Dec 2025

Quantitative Analyst

Toronto, ON

- Built a **Portfolio Optimization & Simulation tool** using historical data and mean-variance optimization to generate optimal Sharpe and minimum-volatility portfolios.
- Developed a **Value-at-Risk tool** using Monte Carlo simulations to assess potential portfolio losses from a distribution of outcomes.
- Adapted an interactive Black-Scholes option pricing calculator to compute call/put prices, implied volatility, and probabilities of expiring in-the-money.

R&R Utility Limited

Sept 2021 – Apr 2024

Project Management Assistant

Toronto, ON

- Supported development of corporate operating budgets, project costing, and drawdown budgets across multi-discipline engineering and construction projects.
- Conducted financial reconciliations on vendor invoices, resolving billing discrepancies and improving cost tracking accuracy.

RESEARCH & PROJECTS

Cluster-Constrained Portfolio Optimization (Undergraduate Thesis) | Python, Scikit-learn

2025 – 2026

- Researched whether sector-clustering constraints improve mean-variance portfolio performance vs. plain MV on a 33-stock sector-stratified universe.
- Built an end-to-end backtesting pipeline: Ward hierarchical clustering → cluster-constrained MV optimization → benchmarking against Plain MV and ERC across 860 rolling 5-year windows.
- Demonstrated cluster-constrained MV consistently outperforms plain MV out-of-sample across weekly and monthly rebalancing horizons.
- Showed plain MV behaves as an *error maximizer* — concentrating allocations in in-sample winners and underperforming out-of-sample.

Anomalies Research — Systematic Equity Strategies | Python, Pandas, NumPy

2025 – Present

- Designed and built the end-to-end backtesting framework and stress-testing pipeline used to evaluate systematic equity anomalies.

- Engineered modular logic for signal generation, portfolio construction, and out-of-sample validation — enabling rapid iteration across new strategy ideas.
- Implemented robustness checks including drawdown analysis, regime testing, and parameter-sensitivity stress tests.

Clustering-Based Selection for Investment Portfolios | *Python, Scikit-learn, PCA, k-means*

Jul 2024

- Built an unsupervised-learning tool to cluster ETFs and strategies by market correlation and risk-adjusted return features; applied PCA + k-means.
- Validated that near-zero correlation between top selected strategies from each cluster held up in out-of-sample testing.

ETF SQL Database Management System | *Python, SQLAlchemy, SQLite, yfinance*

Aug 2024

- Designed a Python tool to manage ETF/stock data via SQLite and SQLAlchemy, with incremental updates that fetch only data past the maximum existing date — reducing API requests and runtime by **over 95%** vs. full reload.

SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, Statsmodels, SQLAlchemy), SQL, MATLAB, R, Java, VBA

Quantitative & Finance: Mean-variance optimization, Monte Carlo simulation, Black-Scholes, Value-at-Risk, backtesting, time-series analysis, clustering, PCA

Tools: Adobe Analytics, Excel (advanced — LAMBDA, dynamic ranges, VBA macros), PowerPoint, Git, Jupyter

ACTIVITIES

Math Tutor (Self-Employed)

Sept 2022 – Present

Private High School Mathematics Instruction

Toronto, ON

- One-on-one tutoring in algebra, calculus, geometry, and trigonometry; built customized lesson plans tailored to individual students' learning styles and academic goals.